Further,

$$a \times b = \frac{-7}{8} \times \frac{3}{5} = \frac{-7 \times 3}{8 \times 5} = \frac{-21}{40}$$

Also, $$b \times a = \frac{3}{5} \times \frac{-7}{8} = \frac{3 \times -7}{5 \times 8} = \frac{-21}{40}$$

Here, we find that $a \times b = b \times a$ and hence multiplication is commutative.

Try these

(i) Is $\frac{3}{5} - \frac{7}{8} = \frac{7}{8} - \frac{3}{5}$?

(ii) Is $\frac{3}{5} \div \frac{7}{8} = \frac{7}{8} \div \frac{5}{3}$?

So, what do you conclude?

1.5.3 Associative property/law for the collection $\mathbb{Q}$ of rational numbers

i) Associative property for Addition

For any three rational numbers ***a***, ***b***, and ***c***, ***a + (b + c) = (a + b) + c***

ii) Associative property for Multiplication

For any three rational numbers ***a***, ***b***, and ***c***, ***a(bc) = (ab)c***

Illustration

Take rational numbers a, b, c as $a = \frac{-1}{2}$, $b = \frac{3}{5}$ and $c = \frac{-7}{10}$

Now, $a + b = \frac{-1}{2} + \frac{3}{5} = \frac{-5}{10} + \frac{6}{10}$ (equivalent rationals with common denominators)

$$a + b = \frac{-5+6}{10} = \frac{1}{10}$$

$$(a+b)+c = \frac{1}{10} + \left(\frac{-7}{10}\right) = \frac{1-7}{10} = \frac{-6}{10} = \frac{-3}{5} \quad \ldots(1)$$

$$\text{Also, } b + c = \frac{3}{5} + \frac{-7}{10} = \frac{6}{10} + \frac{-7}{10} = \frac{6-7}{10} = \frac{-1}{10}$$

$$a+(b+c) = \frac{-1}{2} + \frac{-1}{10} = \frac{-5}{10} + \frac{-1}{10} = \frac{-5-1}{10} = \frac{-6}{10} = \frac{-3}{5} \quad \ldots(2)$$

(1) and (2) shows that $(a+b)+c = a+(b+c)$ is true for rational numbers.

$$\text{Similarly, } a \times b = \frac{-1}{2} \times \frac{3}{5} = \frac{-1 \times 3}{2 \times 5} = \frac{-3}{10}$$

$$(a \times b) \times c = \frac{-3}{10} \times \frac{-7}{10} = \frac{-3 \times -7}{10 \times 10} = \frac{21}{100} \quad \ldots(3)$$

$$\text{Also, } b \times c = \frac{3}{5} \times \frac{-7}{10} = \frac{3 \times -7}{5 \times 10} = \frac{-21}{50}$$

$$a \times (b \times c) = \frac{-1}{2} \times \frac{-21}{50} = \frac{-1 \times -21}{2 \times 50} = \frac{21}{100} \quad \ldots(4)$$

(3) and (4) shows that

$(a \times b) \times c = a \times (b \times c)$ is true for rational numbers. Thus, the associative property is true for addition and multiplication of rational numbers.

Try this

Check whether associative property holds for subtraction and division.

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